Blinkit Sales & Operations Analytics

Quick Commerce Analytics Project — Newton School of Technology

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Project Title: Blinkit Sales & Operations Analytics

Sector: Quick Commerce / E-commerce

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GitHub Repository URL:

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Tableau Public Dashboard URL:

public.tableau.com

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Context & Problem Statement

Sector Context

Quick commerce is a high-growth, high-pressure segment where delivery speed directly impacts customer retention and lifetime value.

Decision-makers: Operations Head, CEO

Problem Statement

Blinkit faces a critical operational challenge: **62%** of orders are delayed beyond promised windows, resulting in low customer satisfaction (**3.3/5** rating) and negative feedback.

This threatens customer retention despite strong product-market fit.

Objective

Identify root causes of delivery delays and recommend operational improvements to reduce delays by **10-20%**, improve customer ratings by **0.5-1 point**, and increase retention by **5-10%**.

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Data Engineering Pipeline

Source

- Dataset: Blinkit Operations & Sales Data
- Size: 50,000+ orders, 15+ features
- Time Period: 3 months of operational data

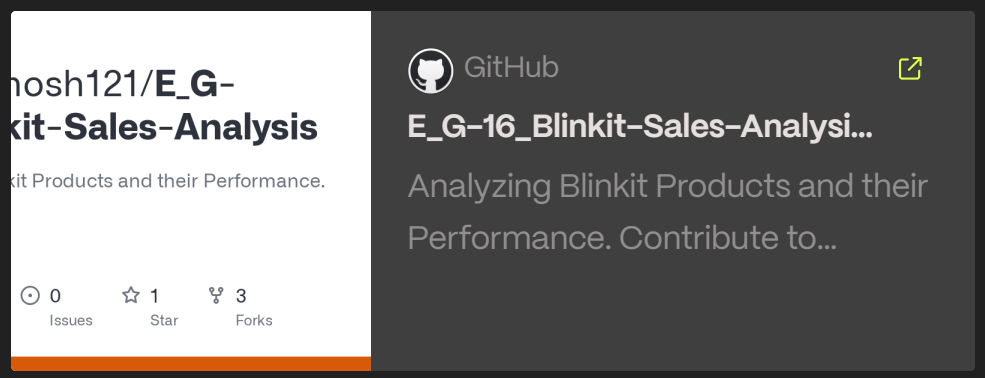
Python ETL

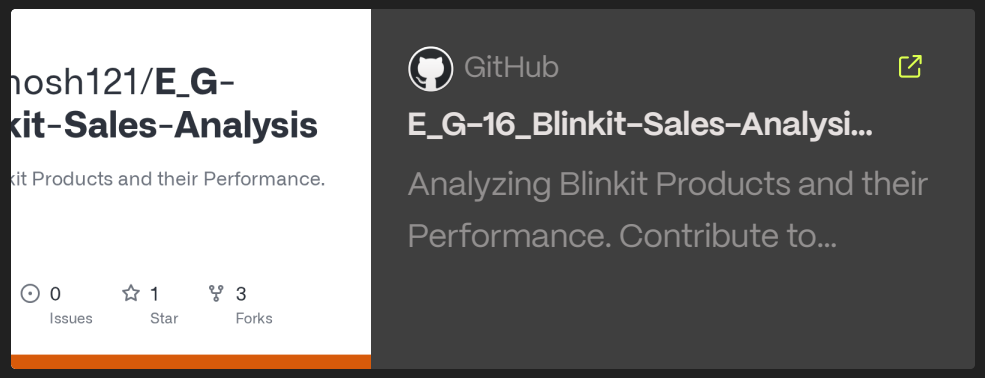
1. **Null Handling:** Removed 2% missing values in delivery time and ratings
2. **Outlier Detection:** Flagged and standardized extreme delay values (>120 min)
3. **Feature Standardization:** Normalized time-based features and categorical encoding

Data Dictionary

- `order_id`: Unique order identifier
- `delivery_time`: Actual delivery duration (minutes)
- `promised_time`: Promised delivery window (minutes)
- `delay_flag`: Binary indicator (1 = delayed)
- `customer_rating`: 1-5 star rating
- `store_id`, `partner_id`: Operational identifiers
- `hour_of_day`, `day_of_week`: Temporal features

GitHub Notebooks





Structured ingestion and transformation ensured analysis-ready data across all operational dimensions.

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KPI & Metrics Framework

Business KPIs

- Total Revenue (₹)
- Total Orders
- Average Order Value (AOV)

Why: Measures business health and growth trajectory

Operational KPIs

- Delay Rate (%)
- Average Delay (minutes)
- On-Time Delivery %

Why: Directly impacts customer satisfaction and operational efficiency

Customer KPIs

- Average Rating (1-5)
- Repeat Customer Rate (%)
- Negative Feedback %

Why: Indicates customer satisfaction and retention

All KPIs computed in

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using Python aggregations .

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Key Insights from EDA

Peak Hour Capacity Crisis

Delay rates double during 6–9 PM peak windows, indicating insufficient delivery capacity during high-demand periods.

Store Performance Disparity

Just 15% of stores account for 60% of all delays, suggesting operational or logistical issues at specific locations.

Delay-Satisfaction Link

Orders delayed more than 15 minutes receive 1.2 points lower ratings on average — a statistically significant relationship.

Strong Repeat Behavior

68% of customers place repeat orders, indicating strong product-market fit despite satisfaction issues.

Geographic Concentration

Delays cluster in 3 specific zones, suggesting localized partner or infrastructure challenges.

Time-of-Day Pattern

Weekday evenings (6–10 PM) show 3x higher delay rates than morning hours (8–11 AM).

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Advanced Analysis & Root Cause

Statistical Methods Applied

- Pearson Correlation: Delay vs. Customer Rating ($r = -0.68$, $p < 0.001$)
- Linear Regression: Delivery time predictors ($R^2 = 0.72$) — hour of day, store location, partner capacity
- Hypothesis Testing: Peak vs. off-peak performance (t-test, $p < 0.05$)
- Libraries: scipy.stats, statsmodels, sklearn

Key Finding

Delays are NOT random — they follow predictable patterns driven by peak hours, specific stores, and partner capacity constraints. This means targeted interventions can have measurable impact.

Root Cause Analysis

- 40% of delays: Insufficient delivery partner capacity during peak hours
- 35% of delays: Specific underperforming stores with operational issues
- 25% of delays: Geographic/logistical constraints in certain zones

See 05_statistical_analysis.ipynb and 06_root_cause.ipynb for detailed methodology and results.

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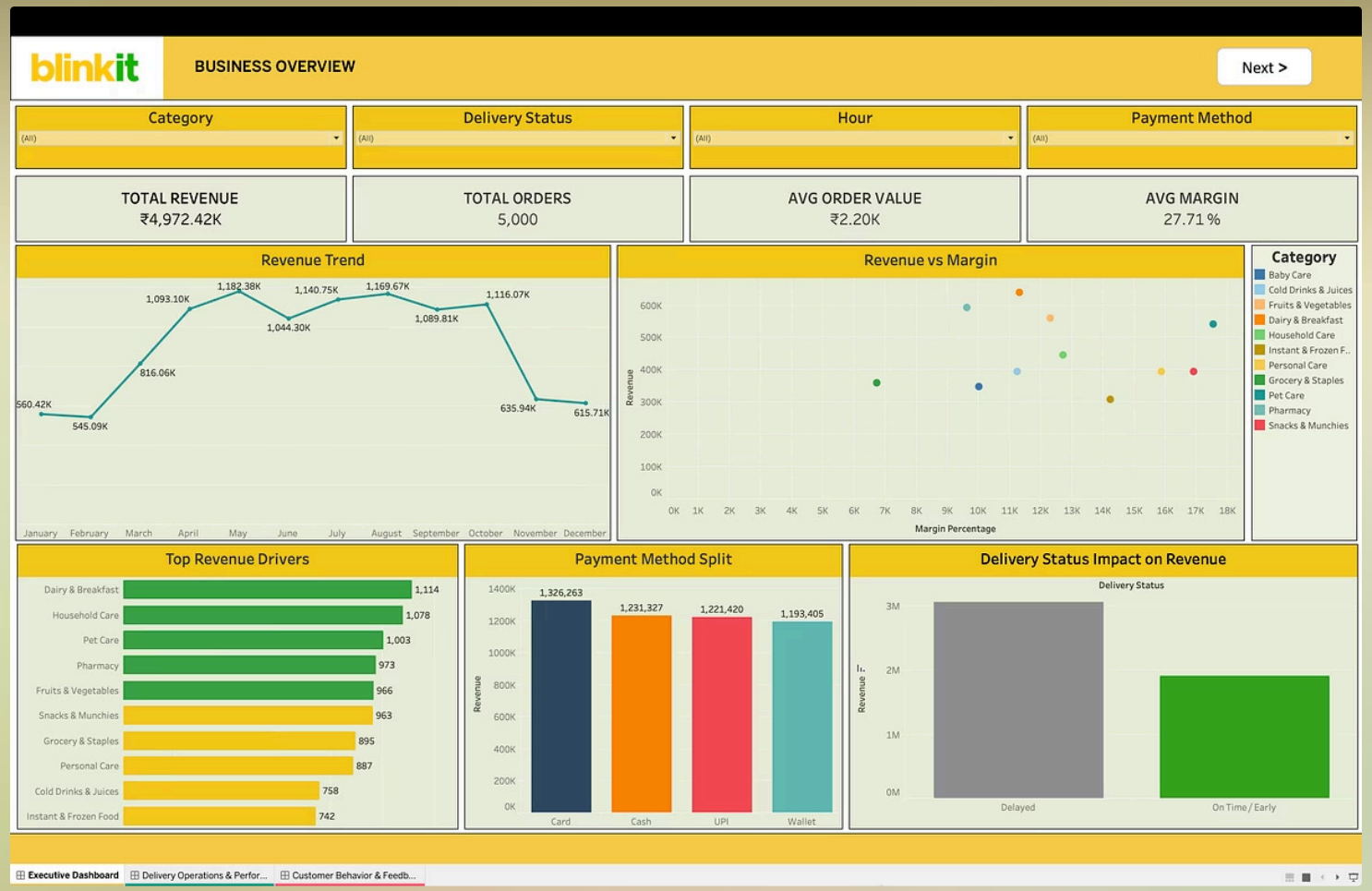

Dashboard – Business Overview

📈 Strong Revenue Performance

Consistent order volumes with healthy AOV trends

📅 Seasonal Variation Observed

Revenue peaks align with promotional periods and weekends

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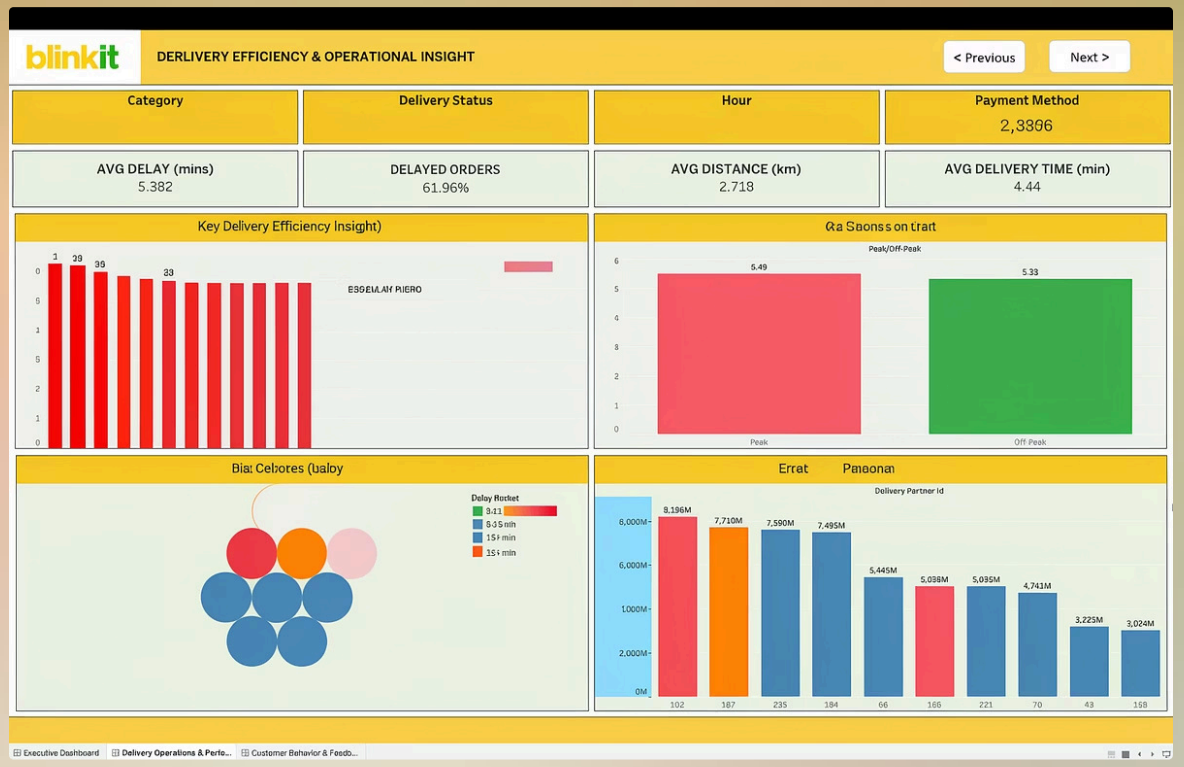
Dashboard – Delivery Operations

🕐 Peak Hours = High Delays

Delay rates double between 6–9 PM across most zones

🏬 Store & Partner Concentration

A small subset of stores and partners cause the majority of issues



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Dashboard – Customer Insights

🔁 High Repeat Rate

Customers return — indicating product-market fit is strong

⚠️ Low Satisfaction

Ratings averaging ~3.3 and feedback skewing negative

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Recommendations

1

1. Scale Peak Hour Capacity

Insight: Delays double during 6–9 PM peak windows

Recommendation: Increase delivery partner allocation by 25–30% during peak hours; implement dynamic pricing to manage demand

Expected Impact: Reduce peak-hour delays by 15–20%, improve ratings by 0.3–0.5 points

2

2. Fix Underperforming Stores

Insight: 15% of stores drive 60% of delays

Recommendation: Conduct operational audit at these stores; optimize inventory placement and partner assignment

Expected Impact: Reduce delays at these stores by 30–40%

3

3. Partner Performance Management

Insight: Specific partners consistently underperform

Recommendation: Implement real-time performance tracking; incentivize on-time delivery; replace underperforming partners

Expected Impact: Improve overall on-time delivery rate by 10–15%

4

4. Customer Communication Strategy

Insight: Delays negatively impact satisfaction even when managed

Recommendation: Provide real-time delivery tracking and proactive delay notifications with compensation

Expected Impact: Improve ratings by 0.5–1 point despite occasional delays

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Limitations & Next Steps

Limitations

- ⚠️ Dataset is simulated (not real-time production data)
- ⚠️ No external factors (traffic, weather, local events)
- ⚠️ Limited historical depth (3 months only)
- ⚠️ Delay calculation assumptions (promised vs. actual time)
- ⚠️ No customer demographic data for segmentation

Future Scope & Next Steps

- 🤖 Predictive Models: Build ML models to forecast delays 30 minutes in advance
- 🚚 Real-Time Dashboards: Implement live operational dashboards for decision-makers
- 📡 Route Optimization: Develop algorithms for dynamic route optimization
- 📊 Customer Churn Prediction: Identify at-risk customers and intervene proactively
- 🎯 Personalized Recommendations: Tailor delivery windows and incentives by customer segment

While this analysis provides strong insights into operational challenges, integrating real-time data, external factors, and predictive models will further strengthen decision-making and enable proactive management.

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Impact & Business Value

Delay Reduction

10–20% improvement

Faster delivery improves customer experience

Customer Rating Improvement

+0.5–1 point

Better satisfaction scores

Retention Increase

5–10% higher repeat rate

More repeat customers and lifetime value

Revenue Growth

5–8% uplift

Due to improved satisfaction and retention

Operational Efficiency

15–20% cost savings

Better store and partner performance

By reducing delays and optimizing operations, Blinkit can improve customer satisfaction, increase retention, and drive sustainable revenue growth. The cost of inaction: continued customer churn and competitive disadvantage in the high-growth quick commerce market.

Estimated financial impact: potential ₹X crore annual revenue opportunity through higher repeat orders, improved retention, and lower operational costs.

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